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A refinement of the Hitchcock definition of actual causation

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Models of actual causation seek to capture the causal relationships between two specific events within a given context, thereby contributing to a clear understanding of causality. A key challenge in constructing such models lies in providing a rigorous and well-defined notion of actual causation. Several influential approaches have been proposed in the literature, including the Hitchcock definition, the modified Halpern-Pearl definition, and the Halpern-Hitchcock definition. However, these definitions struggle to offer a unified solution to certain canonical problems, such as the switch example, the short circuit example, the bogus prevention example, and the switch-variant example. To address this limitation, we propose a definition for actual causation by formalizing and refining the Hitchcock definition. This refined definition offers a framework capable of resolving these canonical problems. Additionally, we present an analysis comparing these definitions, demonstrating that our proposed definition is more restrictive than both the Hitchcock and modified Halpern-Pearl definitions.

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Introduction

Causation is a fundamental concept that plays a critical role across a wide range of disciplines, including artificial intelligence, philosophy, law, biomedicine, economics, and epidemiology, among others. Its importance has inspired extensive interdisciplinary research, as scholars seek to explore its implications and applications.

Within the literature, causation is generally divided into two main categories: type causation (general causation) and actual causation (token causation). Type causation refers to a relationship between types of events in general contexts, whereas actual causation describes a relationship between specific, concrete events within a particular scenario—events tied to particular individuals, places, or times (Hitchcock (2018)).[1] For example, the statement "smoking causes lung cancer" represents type causation, as it captures a general causal relationship. In contrast, the statement "Lucy's smoking ten cigarettes daily for twenty years caused her to develop lung cancer this year" illustrates token causation, as it refers to a specific causal instance involving a particular individual and a definite time frame. Beyond such specific instances, theories of actual causation are crucial. They not only deepen our understanding of the nature of causality but also provide a theoretical framework for addressing practical societal issues, such as responsibility attribution and blame assignment.

To effectively model actual causation, it is crucial to formulate a general definition. In recent years, substantial research efforts have been devoted to addressing this challenge (see, e.g., Beckers (2021); Beckers and Vennekens (2018); Bochman (2018); Hall (2007); Halpern (2008, 2015); Halpern and Hitchcock (2015); Hitchcock (2001); Menzies (2007); Pearl (2000); Weslake (2015); Zhu (2022)). Most definitions of actual causation are rooted in the counterfactual approach introduced by Lewis (1973a), which posits that C is a cause of E if, in the absence of C, E would not have occurred. To resolve the issue of early preemption, Lewis defined causation as the transitive closure of causal dependence among actual events (Lewis (1979)).[2] However, this approach entails the transitivity of causation as a corollary, which has been challenged in the literature on the basis of various counterexamples (Hall (2000)).

One approach to identifying a causal relation without relying on the transitivity principle is to hold certain variables fixed at their actual values, thereby revealing hidden counterfactual dependencies (Hitchcock (2001)). Hitchcock employs explicitly nonforetracking counterfactuals to define an active route between two events. The core idea is to identify an active route between the putative cause and the effect, such that altering the value of the putative cause along this route would also change the value of the effect variable. This approach, referred to as the Hitchcock (HIT) definition, aligns naturally with our intuitive causal judgments by appealing to psychologically natural counterfactuals rather than philosophically technical ones (Hitchcock (2001)). However, the HIT definition struggles to satisfactorily resolve several canonical examples, such as the switch and short circuit problems (Hitchcock (2001)). A similar idea of fixing variables at their actual values is also central to the modified Halpern-Pearl (MHP) definition (Halpern (2015)), which is widely regarded as an influential definition of actual causation in the current literature. Despite its formal elegance, the MHP definition similarly fails to resolve these problems.

A prominent alternative approach involves incorporating notions of normality into accounts of actual causation (Hall (2007); Halpern (2008); Halpern and Hitchcock (2015); Hitchcock (2007); Icard et al. (2017); Menzies (2004, 2007)). This approach is supported by empirical research indicating that people's judgments of actual causation are influenced by their perceptions of the normality of events (see Hitchcock and Knobe (2009)). For instance, Halpern and Hitchcock (2015) introduce a normality ordering over possible worlds to rank causes based on their most normal witness. The Halpern and Hitchcock (HH) definition, which builds on this idea, successfully addresses the short circuit example (Halpern and Hitchcock (2015)). The authors emphasize that the structural equations within the model play a crucial role in determining the normality ordering. Similarly, Weslake (2015) proposes a related approach to handle the switch example, suggesting that the relevant possibilities within the model should be exhaustive. In his framework, there are typically only two possibilities, both leading to the same outcome. Introducing a third possibility that yields a different outcome could lead to confusion and unsatisfactory results. These cases highlight a subtle yet significant constraint that profoundly influences our causal judgments.

Inspired by these insights, we refine the HIT definition by introducing a condition on the notion of active routes and formalizing it within the structural equation framework of Halpern and Pearl (Halpern and Pearl (2005)). Specifically, we consider whether the possible worlds resulting from a hypothetical intervention are consistent with the given context. This refinement yields a relatively simple and elegant definition that provides a unified solution to several canonical problems in the literature.

The remainder of this paper is structured as follows. Section "Basic concepts" introduces the formal framework of structural equations and reviews the HIT, MHP, and HH definitions. Section "A Problem in the HIT and MHP definitions" discusses a key problem faced by both the HIT and MHP definitions. In Section "A refined HIT definition", we refine the HIT definition by incorporating a condition on the notion of active routes and establish a theoretical result regarding the relationships between these definitions. Section "Analyzing other examples" applies the new definition to several canonical examples, comparing its performance with other definitions. Finally, Section "Concluding Remarks" concludes the paper.

Basic concepts

In this section, we introduce key concepts, including causal models and active routes, and provide a review of the HIT, MHP, and HH definitions.

Causal models. Following Halpern and Pearl (2005), we assume that the world is modeled by random variables and their values, with causal influences between these variables represented by a set of structural equations. It is conceptually useful to divide the random variables into two sets: exogenous variables and endogenous variables. To this end, a signature that lists these two types of variables and characterizes their possible values can be defined as follows.

Definition 1. A signature $\mathcal{S}$ is a tuple $(\mathcal{U}, \mathcal{V}, \mathcal{R})$, where $\mathcal{U}$ is a set of exogenous variables, $\mathcal{V}$ is a set of endogenous variables, and $\mathcal{R}$ associates with every variable $Y \in \mathcal{U} \cup \mathcal{V}$ a nonempty set $\mathcal{R}(Y)$ of possible values for Y.

Definition 2. A causal model M is a pair $(\mathcal{S}, \mathcal{F})$, where $\mathcal{S}$ is a *signature*,and $\mathcal{F}$ defines a set of *structural equations*, i.e., $\mathcal{F}$ associates with each endogenous variable $X \in \mathcal{V}$ a function denoted F_X such that $F_X : (\times_{U \in \mathcal{U}} \mathcal{R}(U)) \times (\times_{Y \in \mathcal{V} - \{X\}} \mathcal{R}(Y)) \to \mathcal{R}(X)$.

The structural equations indicate that the value of X is determined by F_X and the values of all other variables in $\mathcal{U} \cup \mathcal{V}$. These equations play a crucial role in defining what happens in the presence of external interventions. Fixing some variable X at x in a causal model $M = (\mathcal{S}, \mathcal{F})$ through an intervention results in a new causal model denoted $M_{X \leftarrow x}$, which is identical to M except

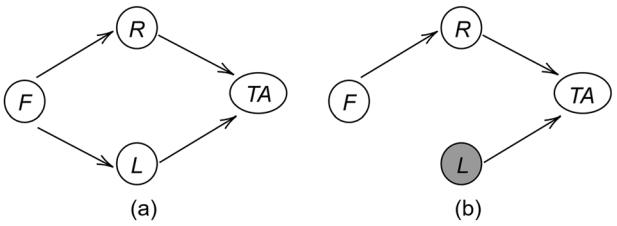


Fig. 1 A graphical representation of a causal model. a Before intervening with variable L. **b** After intervening with variable L.

that the equation for X in $\mathcal{F}$ is replaced by $X = x$.[3] To simplify the exposition, we focus on recursive (or acyclic) models.[4] In such models, for any given context—that is, an assignment $\vec{u}$ to the exogenous variables in $\mathcal{U}$—the system of equations yields a unique solution. A graphical representation of the causal model can often provide useful insight into its structure.

Definition 3. A graph is a profile $\mathcal{G} = (\mathcal{N}, \mathcal{L})$, where $\mathcal{N}$ is a set of nodes (or vertices), and $\mathcal{L}$ is a set of links (or edges) that connect some pairs of vertices.

Each vertex in a graph corresponds to one variable in the model, and the edges denote a certain relation that holds between pairs of variables. If the variable X appears as an argument on the right side of the structural equation for the endogenous variable Z, then X is a parent of Z, and Z is a child of X. There will be an arrow from node X to node Z in this case. The value of each variable is constrained by a function of its parents. A valuation is an assignment of a value to each vertex in accordance with the specified functions. An intervention on X for a valuation is a new valuation with a value of X that violates the function relating X to its parents but is consistent with all other function specifications and agrees with the pre-intervention valuation for each variable that is neither X nor a descendant of X. For example, in Fig. 1(a), F is a parent of R and L, while R and L are parents of TA. We might have $L = F$. Thus, if $F = 0$, then $L = 0$. After intervening with variable L and setting its value to 1, we have $L = 1$ regardless of the value of F. In this case, F is no longer a parent of L (Fig. 1b). To distinguish variables in causal graphs, the variables are shaded when their values are assigned 1.

A route between two variables X and Z is an ordered sequence of variables $(X, \cdots, Z)$ such that each variable in the sequence is a parent of its successor. Graphically, a route between X and Z is a directed path from X to Z. A variable in a path between X and Z is called as an intermediate node of the path. For instance, in Fig. 1(a), there are two paths from F to TA: (F, R, TA) and (F, L, TA). R is an intermediate node of the path (F, R, TA).

Next, we briefly describe a formal language for counterfactuals and interventions, which is useful for defining actual cause properly. For more details, see Halpern (2015).

Definition 4. Given a signature $\mathcal{S} = (\mathcal{U}, \mathcal{V}, \mathcal{R})$, a *primitive event* is a formula of the form $X = x$, where $X \in \mathcal{V}$ and $x \in \mathcal{R}(X)$.

We write $Y \leftarrow y$ to indicate that the value of the variable Y is assigned to y by some external intervention, meaning the structural equation for Y in $\mathcal{F}$ is replaced by $Y = y$.

Definition 5. A *causal formula* over $\mathcal{S}$ is a form $[Y_1 \leftarrow y_1, \cdots, Y_k \leftarrow y_k]\varphi$, where

- φ is a Boolean combination of primitive events;
- $Y_1, \cdots, Y_k$ are distinct variables in $\mathcal{V}$;
- $y_i \in \mathcal{R}(Y_i)$.

A causal formula can be abbreviated as $[\vec{Y} \leftarrow \vec{y}]\varphi$. The special case where $k = 0$ is abbreviated as φ. Intuitively, $[Y_1 \leftarrow y_1, \cdots, Y_k \leftarrow y_k]\varphi$ says that φ would hold if Y_i were set to y_i for $i = 1, \cdots, k$.

A causal formula ψ is true or false in a causal model given a context. As usual, we write $(M, \vec{u}) \models \psi$ if the causal formula ψ is true in the causal model M given the context $\vec{u}$. The $\models$ relation is defined inductively as $(M, \vec{u}) \models (X = x)$ if the variable X has value x in the unique solution to the equations in M with the context $\vec{u}$ (that is, the unique vector of values for the exogenous variables simultaneously satisfies all equations in M with the variables in $\mathcal{U}$ set to $\vec{u}$). The truth of conjunctions and negations is defined in the standard way. Finally, we have

$$(M, \vec{u}) \models [\vec{Y} \leftarrow \vec{y}]\varphi$$

if $(M_{\vec{Y}=\vec{y}}, \vec{u}) \models \varphi$.

Three definitions of actual causation. Inspired by Halpern and Pearl (2005), Hitchcock presents structural equations as tools for representing patterns of counterfactual dependence. He uses explicitly nonforetracking counterfactuals to define the notion of an active route between two events (Hitchcock (2001)). Through this route, the value of the putative cause variable determines the value of the effect variable, indicating a counterfactual dependence between them. If there are multiple routes between the putative cause and the effect, variables on other routes need to be fixed at their actual values. With this idea, actual causation can be identified by finding an active route on the diagram.

Definition 6. A route $(X, Y_1, \cdots, Y_n, Z)$ is active in a causal model $(\mathcal{S}, \mathcal{F})$ if and only if Z depends counterfactually upon X within a new system of equations $\mathcal{F}'$ constructed from $\mathcal{F}$ as follows: for all $Y \in \mathcal{U} \cup \mathcal{V}$, if Y is intermediate between X and Z but does not belong to the route $(X, Y_1, \cdots, Y_n, Z)$, then replace the equation for Y with a new equation that sets Y equal to its actual value in $\mathcal{F}$.

Note that if there is no intermediate variable that does not belong to this route, then $\mathcal{F}'$ is the same as $\mathcal{F}$.

Definition 7. (The HIT definition) Let c and e be distinct occurrent events, and let X and Z be variables such that the values of X and Z represent alterations of c and e, respectively. Then c is a cause of e if and only if there is an active causal route from X to Z in an appropriate causal model $(\mathcal{S}, \mathcal{F})$.

Halpern (2015) offers a definition for actual causation that is elegant and provides a good reference for formalizing our definition. The key idea of this definition is that keeping some variables fixed at their actual values uncovers hidden counterfactual dependence.

Definition 8. (The MHP definition) $\vec{X} = \vec{x}$ is an actual cause of ϕ in $(M, \vec{u})$ if the following three conditions hold:

- AC1. $(M, \vec{u}) \models (\vec{X} = \vec{x})$ and $(M, \vec{u}) \models \phi$.
- AC2(a^m). There is a setting $\vec{W}$ of the variables in $\mathcal{V}$ and a setting $\vec{x}'$ of the variables in $\vec{X}$ such that if $(M, \vec{u}) \models (\vec{W} = \vec{w})$, then $(M, \vec{u}) \models [\vec{W} \leftarrow \vec{w}, \vec{X} \leftarrow \vec{x}']\neg\phi$.
- AC3. $\vec{X}$ is minimal, and no subset of $\vec{X}$ satisfies conditions AC1 and AC2.

It is worth emphasizing that the condition AC2(a^m) is a but-for clause involving a witness. In the context $\vec{u}$, setting $\vec{X} = \vec{x}'$ and $\vec{W} = \vec{w}$ can obtain an assignment of values about the world, meaning that a witness world is one that justifies AC2(a^m). This witnesses the fact that $\vec{X} = \vec{x}$ is a cause of ϕ in context $\vec{u}$.

Halpern and Hitchcock (2015) develop a flexible formal framework for incorporating normality into an account of actual causation. They use a normality ordering on possible worlds to rank causes according to the most normal witness. Before reviewing their definition, let us first provide some notions.

Definition 9. A *world* is an assignment of values to the endogenous variables in a causal model M.

We usually assume that there is a partial preorder $\succcurlyeq$ over the set of worlds, which can be used to compare the normality between different worlds. The formula $w \succcurlyeq w'$ stands for w being at least as normal as w'.

Definition 10. An extended causal model is a tuple $M = (\mathcal{S}, \mathcal{F}, \succcurlyeq)$, where $(\mathcal{S}, \mathcal{F})$ is a causal model and $\succcurlyeq$ is a partial preorder on worlds.

Definition 11. (The HH definition) $\vec{X} = \vec{x}$ is an actual cause of ϕ in an extended causal model M with a context $\vec{u}$ if the following three conditions hold:

- AC1. $(M, \vec{u}) \models (\vec{X} = \vec{x})$ and $(M, \vec{u}) \models \phi$.
- AC2. There is a partition of $\mathcal{V}$ into two subsets $\vec{Z}$, $\vec{W}$ with $\vec{X} \subseteq \vec{Z}$, and there are $\vec{x}'$ and $\vec{w}$ such that

(a) $(M, \vec{u}) \models [\vec{W} \leftarrow \vec{w}, \vec{X} \leftarrow \vec{x}']\neg\phi$, and $w_{\vec{W}=\vec{w},\vec{X}=\vec{x}',\vec{u}} \succcurlyeq w_{\vec{u}}$, where $w_{\vec{u}}$ is the actual world characterized by $(M, \vec{u})$;

(b) $(M, \vec{u}) \models [\vec{W}' \leftarrow \vec{w}, \vec{X} \leftarrow \vec{x}, \vec{Z}' \leftarrow \vec{z}^*]\phi$ for all subsets $\vec{W}'$ of $\vec{W}$ and all subsets $\vec{Z}'$ of $\vec{Z}$ such that $(M, \vec{u}) \models \vec{Z}' = \vec{z}^*$.

- AC3. $\vec{X}$ is minimal, and no subset of $\vec{X}$ satisfies conditions AC1 and AC2.

A problem in the HIT and MHP definitions

Although the HIT and MHP definitions can effectively address the problem of early preemption without invoking the transitivity principle, they fail to adequately explain some canonical examples, such as the switch example (Hall (2007)).

Example 1. (Switch) An engineer is standing by a switch in the railroad tracks. A train approaches in the distance. She flips the switch, causing the train to travel down the right-hand track instead of the left. Since the tracks reconverge ahead, the train arrives at its destination regardless. Is the engineer's flipping of the switch a cause of the train's arrival?

Figure 2 illustrates the causal structure of the example, where *EF* stands for the engineer flipping the switch; *TR* denotes the train traveling on the right-hand track; *TL* denotes the train traveling on the left-hand track; and *TA* represents the train's arrival.

We can construct a causal model $M_S = (\mathcal{U}, \mathcal{V}, \mathcal{R}, \mathcal{F})$ for the switch example, where

- The set of exogenous variables $\mathcal{U} = \{U\}$, where U represents the psychological conditions determining whether the engineer flips the switch. $U = 1$ if the engineer decides to flip the switch, $U = 0$ otherwise. We omit the exogenous variables from the graph.
- The set of endogenous variables $\mathcal{V} = \{EF, TR, TL, TA\}$.
- $\mathcal{R} : \mathcal{V} \to \{1, 0\}$.
- The set of structural equations $\mathcal{F}$ are $TR = EF$, $TL = (1 - EF)$, $TA = max\{TL, TR\}$.

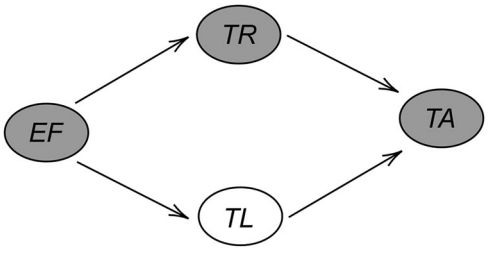


Fig. 2 Switch.

According to the scenario, the actual world is ($EF = 1$, $TR = 1$, $TL = 0$, $TA = 1$). We now analyze the example using the HIT definition. From Fig. 2, there are two paths from *EF* to *TA*: $P_1 = (EF, TR, TA)$, $P_2 = (EF, TL, TA)$. Considering P_1, we fix *TL* at its actual value of 0 and set $EF = 0$, resulting in $TA = 0$. This indicates that *TA* counterfactually depends on *EF*, implying an active route (*EF*, *TR*, *TA*) from *EF* to *TA*. According to the HIT definition, $EF = 1$ is an actual cause of $TA = 1$, which contradicts our intuition.

Next, we analyze the example using the MHP definition. Clearly, AC1 and AC3 hold. We take $\vec{W} = \{TL\}$ and $w = 0$; that is, we hold *TL* at its actual value of 0. If we set $EF = 0$, then $TA = 0$, showing that AC2(a_m) holds. Therefore, we also conclude that $EF = 1$ is an actual cause of $TA = 1$.

However, our intuition suggests that the engineer's flipping of the switch is not an actual cause of the train's arrival. This is because flipping the switch does not actually change the outcome. If the engineer had not flipped the switch, the train would still have arrived at its destination.

One popular approach to addressing this problem is to incorporate the notion of normality into accounts of actual causation. Normality can typically be divided into two types: descriptive (statistical) norms and prescriptive norms (Hitchcock and Knobe (2009)). Descriptive norms are grounded in statistical tendencies, such as the statistical mode or mean, whereas prescriptive norms are rooted in how things ought to be or are supposed to be, encompassing moral, legal, and other normative standards. These norms collectively guide judgments of what constitutes appropriate behavior in a given situation.

The HH definition provides a flexible formal framework for incorporating normality into the account of actual causation. It uses a normality ordering on the set of possible worlds to rank causes according to their most normal witness. Although the HH definition is powerful and can address many problems, it still has limitations, which will be discussed in the next section. Additionally, conflicts may emerge between descriptive and prescriptive norms—an issue even acknowledged by Halpern and Hitchcock themselves (Halpern and Hitchcock (2015)).

A refined HIT definition

In this section, we address the aforementioned problems in the HIT and HH definitions by refining the HIT definition through the addition of a condition on the notion of active routes. Furthermore, we establish a result illustrating the relationships between these definitions.

Although the HH definition is powerful, it encounters difficulties in the *switch-variant example*, where both tracks are blocked (Halpern (2016), p.90).

Example 2. (Switch-variant example) This case builds upon Example 1 by incorporating the possibility of track blockages—either on the right-hand or left-hand side. Figure 3 illustrates the causal structure of this scenario, where *EF* denotes the engineer flipping the switch; *RB* and *LB* indicate blockage states of the right-hand and left-hand tracks, respectively; and *TA* represents the train's arrival outcome. We examine a situation in which both tracks are blocked: the engineer flips the switch, directing the train to the right-hand track instead of the left. Since both tracks are obstructed, the train cannot reach its destination via either route. This analysis aims to determine whether the engineer's action of flipping the switch constitutes a cause of the train's failure to arrive.

Observation 1: In the switch-variant example, we have

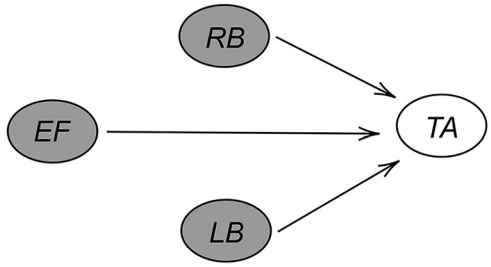


Fig. 3 A variant of the switch example.

(a) Given that the normal values of *EF*, *LB* and *RB* are 0, flipping the switch ($EF = 1$) is a cause of the train not arriving ($TA = 0$) under the HH definition;
(b) $EF = 1$ is not a cause of $TA = 0$ under the MHP definition;
(c) $EF = 1$ is not a cause of $TA = 0$ under the HIT definition.

Proof. We construct a causal model $M'_S = (\mathcal{U}, \mathcal{V}, \mathcal{R}, \mathcal{F})$ for the switch-variant example, where

- $\mathcal{U} = \{U\}$, with $\mathcal{U} = \{U_1, U_2, U_3\}$, where U_1 represents the conditions determining whether the engineer flips the switch: U_1=1 if he decides to flip the switch, and U_1=0 otherwise; U_2 represents the conditions determining whether the left-hand track is blocked: U_2=1 if it is blocked, and U_2=0 otherwise; U_3 represents the conditions determining whether the right-hand track is blocked: U_3=1 if it is blocked, and U_3=0 otherwise.
- $\mathcal{V} = \{EF, RB, LB, TA\}$.
- $\mathcal{R} : \mathcal{V} \to \{1, 0\}$.
- $\mathcal{F} = \{TA = ((EF \wedge \neg RB) \vee (\neg EF \wedge \neg LB))\}$, where $\neg$ *EF* denotes the engineer not flipping the switch, and $\neg$ *RB* and $\neg$ *LB* represent the right-hand and left-hand tracks not being blocked, respectively.

Given the context where both tracks are blocked, the actual world is ($EF = 1$, $LB = 1$, $RB = 1$, $TA = 0$). For part (a), let $\vec{W} = \{LB, RB\}$, and $w = \{0, 1\}$. Then we have

$$(M, \vec{u}) \models [LB \leftarrow 0, RB \leftarrow 1, EF \leftarrow 0](TA = 1).$$

On the assumption that the normal values of *EF*, *LB* and *RB* are 0, the witness world ($EF = 0$, $LB = 0$, $RB = 1$, $TA = 1$) is more normal than the actual world ($EF = 1$, $LB = 1$, $RB = 1$, $TA = 0$). Thus, AC2(a) does hold, and it follows that $EF = 1$ is a cause of $TA = 0$ under the HH definition.

For part (b), let $\vec{W} = \{LB, RB\}$, and $w = \{1, 1\}$. Then we have

$$(M, \vec{u}) \models [LB \leftarrow 1, RB \leftarrow 1, EF \leftarrow 0](TA = 0).$$

This implies that AC2(a^m) does not hold. Hence, $EF = 1$ is not a cause of $TA = 0$ under the MHP definition.

For part (c), consider the path (*EF*, *TA*). Assign actual values of *LB* and *RB* to 1, and set $EF = 0$; then the value $TA = 0$. This indicates that *TA* does not counterfactually depend upon *EF*, which implies that the path (*EF*, *TA*) is not an active route. By the HIT definition, $EF = 1$ is not an actual cause of $TA = 0$. □

In this example, since both tracks are blocked, the train cannot possibly reach its destination. Our intuition is clear: the engineer's act of flipping the switch does not cause the train's failure to arrive. Both the MHP and HIT definitions yield conclusions consistent with this intuition, whereas the HH definition encounters difficulties in this scenario. In our view, the HH definition is overly permissive, as it allows variables outside the causal pathway to be fixed at non-actual values. By considering a broader set of counterfactual worlds, this approach increases the likelihood of classifying an event as a cause. Although the HH definition incorporates normality constraints to restrict contingencies, it remains prone to problems when the actual world contains numerous abnormal endogenous variables. In this example, the witness world ($EF = 0$, $LB = 0$, $RB = 1$, $TA = 1$) is more normal than the actual world ($EF = 1$, $LB = 1$, $RB = 1$, $TA = 0$), since *EF* and *LB* take more normal values in the witness world and *RB* remains unchanged. As a result, the definition inappropriately identifies the engineer's flipping of the switch as a cause.

Definition 12. (Availability) A world is said to be available in a causal model if it does not violate the structural equations of the model.

Essentially, an available world in a causal model is one that is consistent with the given scenario. For example, in the switch example, there are 16 possible worlds, as there are four elements in the set of endogenous variables. The structural equations in this example are defined as $TR = EF$, $TL = (1 - TR)$, $TA = max\{TL, TR\}$. According to the scenario, if $EF = 1$, then $TR = 1$, $TL = 0$, and $TA = 1$. Conversely, if $EF = 0$, then $TR = 0$, $TL = 1$, and $TA = 1$. Therefore, the world ($EF = 1$, $TR = 1$, $TL = 0$, $TA = 1$) and ($EF = 0$, $TR = 0$, $TL = 1$, $TA = 1$) are available in the model M_S. In contrast, the other worlds are not available in the model, because they violate the structural equations.

As previously established, flipping the switch does not qualify as a cause of the train's arrival in the standard switch example. However, it would be deemed a cause if we account for the possibility that the tracks might be blocked. Thus, when the specific details of a scenario change, the corresponding models and conclusions shift accordingly. From this perspective, the structural equations that constrain the notion of availability are contingent. Nevertheless, studies of actual causation focus on exploring causal relationships within a specific scenario, where the causal influences between events are modeled through a set of structural equations. Once a specific narrative context is provided and background information is fully supplemented, a causal model can be derived; in this case, the structural equations are also determined.

We return to the HIT definition, which is more natural and only fixes variables at their actual values. In the switch example, Weslake (2015) notes that the relevant possibilities should be exhaustive. As we can see, there are only two available worlds in this example: ($EF = 1$, $TR = 1$, $TL = 0$, $TA = 1$) and ($EF = 0$, $TR = 0$, $TL = 1$, $TA = 1$), both of which yield the same result. It is therefore reasonable to conclude that $EF = 1$ is not an actual cause of $TA = 1$. If the world ($EF = 0$, $TR = 0$, $TL = 0$, $TA = 0$) were available, flipping the switch could be considered as a cause —but this is not the case in the switch example. Evidently, the notion of availability helps address the switch example.

The idea that possible worlds should be available in a causal model can also address the short circuit example (Hitchcock (2007)). In this example, there are two possible worlds, both yielding the same result. Consequently, nothing is considered a cause (more details about this example will be discussed in Section 5). Beckers and Vennekens (2018) argue that this example is essentially no different from the switch example. Halpern and Hitchcock (2015) contend that the structural equations in the model significantly influence the normality ordering. Halpern (2015) further discusses that the witness is an abnormal world that contradicts the story.

There appears to be a *subtle restriction* that significantly influences our causal judgments. We now refine the HIT definition with this idea to address the problems it encounters. Specifically, we make important modifications to the notion of active routes. In the refined HIT definition of actual causation we will offer, $X = x$ is an actual cause of $Z = z$ if there is an active causal

route P from X to Z in an appropriate causal model M. The route $P = (X, Y_1, \cdots, Y_n, Z)$ is active in $(M, \vec{u})$ if and only if Z depends counterfactually upon X within the new equations $\mathcal{F}'$ constructed from $\mathcal{F}$ as follows: for the intermediate $\vec{W}_{xz}$ between X and Z, if $W \in \vec{W}_{xz}$, but does not belong to the route P, then replace the equation for W with a new equation that sets W equal to its actual value w in $\mathcal{F}$. Crucially, the possible world $(x', \vec{w}_{xz}, y'_1, \cdots, y'_n, z')$ produced by intervention must be available in the model. As Woodward (2016) argues, this constraint should be incorporated into the formulation of the given problem, and it is assumed to be inviolable.

With the concept of availability, a refined HIT (RHIT) definition of actual causation can be formulated as follows:

Definition 13. (An RHIT definition) $X = x$ is an actual cause of $Z = z$ in $(M, \vec{u})$ if the following conditions hold:

- AC1. $(M, \vec{u}) \models (X = x)$ and $(M, \vec{u}) \models (Z = z)$.
- AC2. In the graph $\mathcal{G}$ characterized by $(M, \vec{u})$, there is a route $P = (X, Y_1, \cdots, Y_n, Z)$ from X to Z and $\vec{W}_{xz}$ is the set of the intermediate nodes between X and Z in which every node is not in the route P. There exists a setting x' of the variable X such that if $(M, \vec{u}) \models (\vec{W}_{xz} = \vec{w}_{xz})$, then

$$(M, \vec{u}) \models [\vec{W}_{xz} \leftarrow \vec{w}_{xz}, X \leftarrow x'](Z = z')$$

and the world $(x', \vec{w}_{xz}, y'_1, \cdots, y'_n, z')$ is available in the model.

Next, we explore the connections between definitions for actual causation.

Lemma 1. (Halpern (2016)) If $(M, \vec{u}) \models [\vec{X} \leftarrow \vec{x}](\vec{Y} = \vec{y})$, then $(M, \vec{u}) \models [\vec{X} \leftarrow \vec{x}]\varphi$ if and only if $(M, \vec{u}) \models [\vec{X} \leftarrow \vec{x}, \vec{Y} \leftarrow \vec{y}]\varphi$.

Proposition 1. Let M be a causal model and $\vec{u}$ be a context, then

(a) If $X = x$ is a cause of $Z = z$ in $(M, \vec{u})$ under the RHIT definition, then $X = x$ is a cause of $Z = z$ in $(M, \vec{u})$ under the HIT definition.
(b) If $X = x$ is a cause of $Z = z$ in $(M, \vec{u})$ under the HIT definition, then $X = x$ is a cause of $Z = z$ in $(M, \vec{u})$ under the MHP definition.
(c) If $X = x$ is a cause of $Z = z$ in $(M, \vec{u})$ under the RHIT definition, and for every available w, $w \overline{\succ} w_{\vec{u}}$, then $X = x$ is a cause of $Z = z$ in $(M, \vec{u})$ under the HH definition.

Proof. The difference between the RHIT and HIT definitions lies in the availability condition, which requires that the world $(x', \vec{w}_{xz}, y'_1, \cdots, y'_n, z')$ is available in the model. It is straightforward to see that part (a) holds.

Suppose that $X = x$ is a cause of $Z = z$ in $(M, \vec{u})$ under the HIT definition. By this definition, there must exist a value x' of the variable X and an intermediate set $\vec{W}_{xz}$ such that if $(M, \vec{u}) \models (\vec{W}_{xz} = \vec{w}_{xz})$, then $(M, \vec{u}) \models [\vec{W}_{xz} \leftarrow \vec{w}_{xz}, X \leftarrow x'](Z = z')$. Clearly, we have $(M, \vec{u}) \models (X = x) \wedge (Z = z)$, since $X = x$ is a cause of $Z = z$ in $(M, \vec{u})$ under the HIT definition. Hence, AC1 holds. Since $(M, \vec{u}) \models [\vec{W}_{xz} \leftarrow \vec{w}_{xz}, X \leftarrow x'](Z = z')$, the obvious candidate for a witness for AC2(a_m) is $(\vec{W}_{xz}, \vec{w}_{xz}, x')$. AC3 holds for $X = x$, as it is a singleton. Thus part (b) holds.

Finally, assume that $X = x$ is a cause of $Z = z$ in $(M, \vec{u})$ under the RHIT definition. The proofs of AC1 and AC3 in the HH definition are similar to those of part (b). We need to prove that AC2(a) and AC2(b) hold. Since $X = x$ is a cause of $Z = z$ in $(M, \vec{u})$, there must exist a value x' of the variable X and an intermediate set $\vec{W}_{xz}$ such that if $(M, \vec{u}) \models (\vec{W}_{xz} = \vec{w}_{xz})$, then $(M, \vec{u}) \models [\vec{W}_{xz} \leftarrow \vec{w}_{xz}, X \leftarrow x'](Z = z')$. The obvious candidate for a witness for AC2(a) is $(\vec{W}_{xz}, \vec{w}_{xz}, x')$. By the RHIT definition, the world $(x', \vec{w}_{xz}, y'_1, \cdots, y'_n, z')$ is available in the model. For every available w, $w \overline{\succ} w_{\vec{u}}$, by the assumption $w \overline{\succ} w_{\vec{u}}$, we have $w_{\vec{W}_{xz} = \vec{w}_{xz}, X = x', \vec{u}} \overline{\succ} w_{\vec{u}}$. Hence, AC2(a) holds.

According to the RHIT definition, we have $(M, \vec{u}) \models (X = x) \wedge (Z = z)$, $(M, \vec{u}) \models (\vec{W}_{xz} = \vec{w}_{xz})$. By the HH definition, we have $(M, \vec{u}) \models (\vec{T}' = \vec{t}^*)$. It is straightforward to show that $(M, \vec{u}) \models [\vec{W}_{xz} \leftarrow \vec{w}_{xz}, X \leftarrow x](\vec{T}' = \vec{t}^*)$ and $(M, \vec{u}) \models [\vec{W}_{xz} \leftarrow \vec{w}_{xz}, X \leftarrow x](Z = z)$. By Lemma 1, we have $(M, \vec{u}) \models [\vec{W}_{xz} \leftarrow \vec{w}_{xz}, X \leftarrow x, \vec{T}' \leftarrow \vec{t}^*](Z = z)$. It follows that AC2(b) holds. □

Parts (a) and (b) of Proposition 1 show that the MHP definition is the most permissive of the three, while the RHIT definition is the most restrictive. It should be noted that the converse of (a) does not hold. A counterexample can be found in the switch example, where flipping the switch is a cause of the train's arrival under the HIT definition but not under the RHIT definition. Halpern (2015) provides an example illustrating that the converse of (b) also does not hold.[5]

Observation 2: For the switch-variant example, flipping the switch ($EF = 1$) is not a cause of the train not arriving ($TA = 0$) under the RHIT definition;

Proof. Let $\vec{W}_{xz} = \{LB, RB\}$, and $w_{xz} = \{1, 1\}$. Then, we have $(M, \vec{u}) \models [LB \leftarrow 1, RB \leftarrow 1, EF \leftarrow 0](TA = 0)$. Thus, AC2 is not satisfied, which implies this conclusion. □

Analyzing other examples

In this section, we analyze some well-known examples using the RHIT, HH, MHP, and HIT definitions. We first consider the short circuit example (Hitchcock (2007)).

Example 3. (Short Circuit) The Assistant Bodyguard administers a harmless antidote to Victim's coffee. Subsequently, Buddy poisons the coffee with a substance that is ordinarily lethal—though this toxicity is neutralized by the antidote. Critically, Buddy would not have poisoned the coffee at all had Bodyguard not first administered the antidote. Victim consumes the coffee and survives. A question arises: Does Bodyguard's act of administering the antidote constitute a cause of Victim's survival?

The corresponding causal graph for this example is shown in Fig. 4, where A represents Buddy putting poison in the coffee; B denotes the bodyguard putting the antidote in the coffee; and VS represents the victim's survival.

Observation 3: The following assertions hold in the short circuit example.

(a) Putting in the antidote ($B = 1$) is not a cause of the victim's survival ($VS = 1$) under the RHIT definition;
(b) $B = 1$ is not a cause of $VS = 1$ under the HH definition;
(c) $B = 1$ is a cause of $VS = 1$ under the MHP definition;
(d) $B = 1$ is a cause of $VS = 1$ under the HIT definition.

Proof. We construct a causal model $M_{SC} = (\mathcal{U}, \mathcal{V}, \mathcal{R}, \mathcal{F})$ for the short circuit example, where

- $\mathcal{U} = \{U_1, U_2\}$, where U_1 represents the conditions determining whether Buddy puts poison in the coffee: $U_1 = 1$ if he decides to put the poison, and $U_1 = 0$ otherwise; U_2 represents the conditions determining whether the bodyguard puts the antidote in the coffee: $U_2 = 1$ if he decides to put the antidote, and $U_2 = 0$ otherwise.

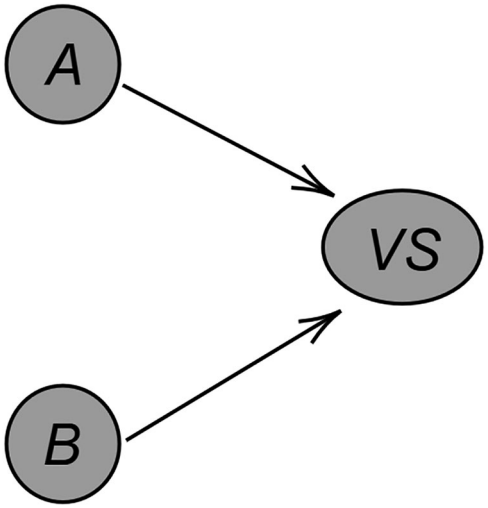


Fig. 4 Short Circuit.

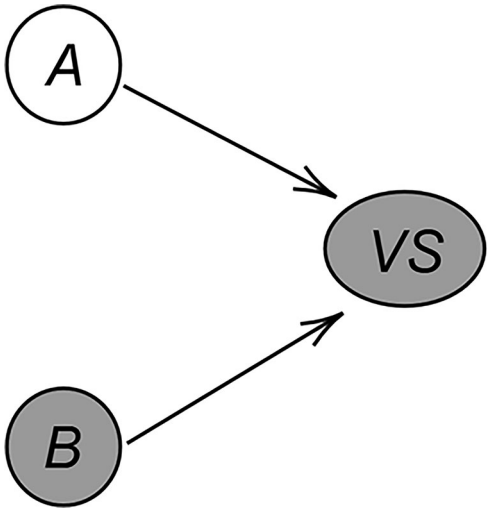


Fig. 5 Bogus Prevention.

- $\mathcal{V} = \{A, B, VS\}$.
- $\mathcal{R} : \mathcal{V} \rightarrow \{1, 0\}$.
- $\mathcal{F}$ includes $A = B$, $VS = max\{(1 - A), B\}$.

In the actual world, we have $A = 1$, $B = 1$, and $VS = 1$, so AC1 hold in these definitions. AC3 trivially holds for $B = 1$, as it is a singleton. Figure 4 shows that there is a path (B, VS) with the intermediate variable A. Let $\vec{W}_{xz} = \{A\}$ and $w_{xz} = 1$. Then, we have $(M, \vec{u}) \models [A \leftarrow 1, B \leftarrow 0](VS = 0)$. Note that the world ($A = 1$, $B = 1$, $VS = 1$) and ($A = 0$, $B = 0$, $VS = 1$) are available within the model, whereas the world ($A = 1$, $B = 0$, $VS = 0$) is not. Thus, AC2 does not hold, which implies that $B = 1$ is not a cause of $VS = 1$ under the RHIT definition. Therefore, part (a) holds.

For part (b), let $\vec{W} = \{A\}$ and $w = 1$. Then, we have $(M, \vec{u}) \models [A \leftarrow 1, B \leftarrow 0](VS = 0)$. The witness world ($A = 1$, $B = 0$, $VS = 0$) directly contradicts the story, meaning that it is less normal than the actual world ($A = 1$, $B = 1$, $VS = 1$). Thus, AC2(a) does not hold, and $B = 1$ is not a cause of $VS = 1$ under the HH definition.

For part (c), let $\vec{W} = \{A\}$ and $w = 1$. Then, we have $(M, \vec{u}) \models [A \leftarrow 1, B \leftarrow 0](VS = 0)$, which implies that AC2(a^m) does hold. Thus, $B = 1$ is a cause of $VS = 1$ under the MHP definition.

For part (d), consider the path (B, VS). Assign the actual value of A to 1, and set $B = 0$; then the value $VS = 0$. This means the path (B, VS) is an active route. Thus, $B = 1$ is a cause of $VS = 1$ under the HIT definition. □

From an intuitive standpoint, the majority of people would concur that administering the antidote does not constitute a cause of Victim's survival. Instead, Bodyguard and his associate Buddy appear to be playing a deliberate trick on Victim—one that aligns with Hitchcock's speculative scenario: we might suppose that Bodyguard, who is up for a promotion (the bodyguard equivalent of tenure), wants to make it look as though he has foiled an assassination attempt, with Buddy assisting him (Hitchcock (2007)). Critically, their motives were not to inflict harm on Victim; rather, the ruse was designed to fabricate a "life-saving" scenario. Further, the act of administering the antidote did not produce any substantive change in the final outcome: had Bodyguard refrained from adding the antidote, Buddy would not have poisoned the coffee at all, and Victim would still have survived by drinking the unadulterated beverage. For these reasons, we take the view that Bodyguard's administration of the antidote is not a cause of Victim's survival.

The next example is the bogus prevention example (Hitchcock (2007)).

Example 4. (Bogus Prevention) An assassin is in possession of a lethal poison but has a last-minute change of heart and refrains from putting it in the victim's coffee. A bodyguard puts an antidote in the coffee, which would have neutralized the poison had there been any. The victim drinks the coffee and survives. Is the bodyguard's putting in the antidote a cause of the victim's survival?

See Fig. 5 for the causal structure in the example, where A denotes the assassin putting poison in the coffee; B stands for that the bodyguard putting the antidote in the coffee; and VS represents the victim's survival.

Observation 4: In the bogus prevention example, the following assertions hold.

(a) Putting in the antidote ($B = 1$) is not a cause of the victim's survival ($VS = 1$) under the RHIT definition;
(b) Given that the normal values of both A and B are 0, $B = 1$ is not a cause of $VS = 1$ under the HH definition;
(c) $A = 0 \wedge B = 1$ is a cause of $VS = 1$ under the MHP definition ($B = 1$ is part of a cause of $VS = 1$);
(d) $B = 1$ is not a cause of $VS = 1$ under the HIT definition.

Proof. We establish a causal model $M_{BP} = (\mathcal{U}, \mathcal{V}, \mathcal{R}, \mathcal{F})$ for the bogus prevention example, where

- $\mathcal{U} = \{U_1, U_2\}$, where U_1 represents the conditions determining whether the assassin puts poison in the coffee: $U_1 = 1$ if he decides to put the poison, and $U_1 = 0$ otherwise; U_2 represents the conditions determining whether the bodyguard puts the antidote in the coffee: $U_2 = 1$ if he decides to put the antidote, and $U_2 = 0$ otherwise.
- $\mathcal{V} = \{A, B, VS\}$.
- $\mathcal{R} : \mathcal{V} \rightarrow \{1, 0\}$.
- $\mathcal{F} = \{VS = max\{(1 - A), B\}\}$.

In the actual world of this example, we have ($A = 0$, $B = 1$, $VS = 1$), so AC1 holds in the three definitions. AC3 clearly holds for $B = 1$, since it is a singleton.

For part (a), Fig. 5 shows there is a path (B, VS), where the intermediate variable is A. Let $\vec{W}_{xz} = \{A\}$ and $w_{xz} = 0$. Then, we have $(M, \vec{u}) \models [A \leftarrow 0, B \leftarrow 0](VS = 1)$, which entails that AC2

Table 1 Results of Applying the Proposed Definitions to Respective Examples (Symbol Interpretation: ○ = Example Resolved Successfully; △ = Issues in Example Resolution).

	Switch	Short Circuit	Bogus Prevention	Switch-variant
HIT	△	△	○	○
MHP	△	△	△	○
HH	○	○	○	△
RHIT	○	○	○	○

does not hold. Hence, $B = 1$ is not a cause of $VS = 1$ under the RHIT definition.

For part (b), let $\vec{W} = \{A\}$ and $w = 1$. Then, we have $(M, \vec{u}) \models [A \leftarrow 1, B \leftarrow 0](VS = 0)$. On the assumption that the normal values of both A and B are 0, the witness world ($A = 1$, $B = 0$, $VS = 0$) and the actual world ($A = 0$, $B = 1$, $VS = 1$) are incomparable. Thus, AC2(a) does not hold, which means that $B = 1$ is not a cause of $VS = 1$ under the HH definition.

For part (c), let $\vec{W} = \varnothing$. Then, we have $(M, \vec{u}) \models [A \leftarrow 1, B \leftarrow 0](VS = 0)$. Hence, AC2($a^m$) holds, and thus $A = 0 \wedge B = 1$ is a cause of $VS = 1$ under the MHP definition.

For part (d), consider the path (B, VS). Assign the actual value of A to 0, and set $B = 0$; then the value $VS = 1$. This implies that the path (B, VS) is not an active route. Thus, $B = 1$ is not a cause of $VS = 1$ under the HIT definition. □

In summary, the results of the above observations are shown in Table 1, where we see that the RHIT definition provides a treatment of all these canonical problems.

Concluding remarks

Providing a precise definition of causation is a fundamental step in constructing models for actual causation. The HIT, MHP, and HH definitions offer valuable frameworks for determining causal relationships between specific events. However, because the possible worlds resulting from a hypothetical intervention may be inconsistent with the given context, these definitions sometimes produce unsatisfactory causal judgments in certain situations. Consequently, they fail to provide a uniform solution to some canonical problems, such as the switch example, the short circuit example, the bogus prevention example, and the switch-variant example.

To address this problem in these causal theories, we propose a refined definition of actual causation by formalizing and refining the HIT definition. This refined definition provides a treatment of all the aforementioned canonical examples. Our key insight is that certain subtle restrictions, which significantly influence causal judgments, must be incorporated into the modeling of actual causation. Specifically, it is essential to determine whether the possible worlds generated by hypothetical interventions are consistent with a given causal context. By integrating this restriction into the causal models, we propose an alternative approach to determining causal relationships.

To be sure, this restriction might give rise to undesired consequences. For instance, the RHIT definition entails that the trainee's firing does not count as a cause of the victim's death in the early preemption scenario. Nevertheless, we can resolve this counterintuitive conclusion by drawing on Beckers and Vennekens' analysis of the early preemption example (Beckers and Vennekens (2017, 2018)). In their work, they argue that the key distinction between the early preemption and the switch example lies in whether we take into account the possibility that the backup mechanism malfunctions. More specifically, since it is quite plausible to regard the supervisor's shooting accuracy as far more uncertain than the breakage of a sturdy track, it is reasonable to expect that intuitions supporting the trainee's firing as a cause are more prevalent than those endorsing the flip of the switch as a cause. Hence, an appropriate model for the early preemption scenario should incorporate a variable representing the uncertainty inherent in the counterfactual narrative, which stands in contrast to the model constructed for the switch example (Beckers and Vennekens (2018)).

From this perspective, a proper formal model of the early preemption scenario should be nondeterministic—reasonably allowing for the possibility that even the supervisor may not always fire accurately or might similarly lose his nerve. This idea supports the view that our intuitive judgments do not treat the early preemption example as a deterministic case, and that the causal model should accordingly reflect this nondeterminism. To capture the supervisor's potential inaccuracy or hesitation, we can introduce additional variables such as *Accurate* (representing whether the supervisor's firing hits the target) and *Nerve* (representing whether the supervisor maintains the resolve to fire). Integrating these variables into the model yields a more adequate, nondeterministic formalization, which allows us to conclude that the trainee's firing is a cause of the victim's death.

An alternative approach to representing the restriction in causal models can be based on similarity semantics. Lewis (1973b) proposed a method for evaluating counterfactuals that incorporates the notion of comparative similarity: one orders possible worlds according to their distance from the actual world, and the counterfactual "If it were A, then B" is declared true in a world w if and only if B is true in all the closest A-worlds to w. Lewis (1979) subsequently established an intricate system of weights and priorities for various aspects of similarity, aiming to align similarity more closely with causal intuitions. It is for this reason that the framework of similarity semantics offers an alternative approach to interpreting the specific meaning of normality. Furthermore, similarity semantics may provide a feasible method for refining the conditions in the definition of availability. Drawing on insights from this semantics to establish a set of normative assumptions that select a witness world yielding the desired conclusion thus constitutes a promising avenue for future research—one that we intend to explore in our subsequent work.

Data availability

No datasets were generated or analysed during the current study.

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Notes

1 The term "particular individual" denotes a specific entity that can be uniquely and exclusively identified within a defined scope, set of conditions, or given context. While most commonly referring to a human being, it may also denote an animal or inanimate object. Its core feature is that, within a set of similar entities, the target entity is uniquely pinpointed and thereby distinguished from all other members of that set by virtue of satisfying predefined restrictive criteria.

2 The early preemption scenario is as follows. An assassin-in-training is on his first mission. When the supervisor observes the trainee firing the shot, the supervisor refrains from shooting. As a result, the trainee fires his gun and kills the victim. However, if the trainee had not fired, the supervisor would have shot and killed the victim in the same manner. In this case, we regard the trainee's shooting as the cause of the victim's death, even though there is no causal dependence between the trainee's shooting and the victim's death: had the trainee not fired, the supervisor would have acted, and the victim would still have died.

3 For a detailed account of interventions, please refer to Woodward (2004).

4 Here, we restrict our discussion to structural equation models, where "recursive" and "acyclic" are strictly synonymous. Within the terminological framework of structural equation models, recursion denotes a partial ordering of variables, from which a directed acyclic graph is derived. In the broader vocabulary of computer science,

however, recursion can imply the presence of a cycle—recursive functions being a case in point.

5 See Example 4.2 in Halpern (2015).

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The authors declare no competing interests.

Ethical approval

Ethical approval was not required for this study, as it did not involve human participants. No primary surveys were conducted.

Informed consent

This article does not contain any studies with human participants performed by any of the authors.

Additional information

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